

Nafiul Islam

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Summary

Software Engineer with **6+ years of experience** specializing in backend development using **Java, Spring Boot, and Microservices**, with hands-on experience designing scalable systems, RESTful APIs, and high-performance database architectures. Proven track record building production-grade platforms across **IoT, healthcare, travel, aviation, and real-time communication** domains. Skilled in **Python (FastAPI), ReactJS**, and modern **AI-enabled features** using LangGraph multi-agent orchestration, Spring AI, and Retrieval-Augmented Generation (RAG). Experienced in **mentoring junior engineers** and collaborating with cross-functional teams to deliver reliable, maintainable software at scale.

Technical Skills

Programming Languages: Java, Python, JavaScript (ES6+), C, C++, SQL

Backend & Frameworks: Spring Boot, Spring Security, Spring Data JPA, Hibernate, Microservices, Javalin, FastAPI, WebSocket, Apache Kafka, Redis, Swagger/OpenAPI, JWT, OAuth 2.0, JUnit, Mockito

Frontend & Web: ReactJS, HTML5, CSS3, JavaScript, Responsive Design, Real-Time UIs (WebSocket)

AI & Intelligent Systems: LangGraph (multi-agent), LangChain, Spring AI, Retrieval-Augmented Generation (RAG), pgvector, ONNX embeddings, Groq LLaMA 3.3, OpenAI/Claude APIs, prompt engineering

Databases: PostgreSQL, MySQL, MS SQL, SQLite, InfluxDB, pgvector

Cloud & DevOps: Docker, Kubernetes, AWS (EC2, S3, CloudFront, Cognito), CI/CD (GitHub Actions), ELK Stack

Tools & Methodologies: GitLab, GitHub, Bitbucket, Jira, ClickUp, Maven, Gradle, Agile/Scrum, Code Review, TDD, REST API Design, System Design

Work Experience

Cloudly Infotech Limited, Dhaka

January 2025 – Present

Senior Software Engineer

Project: Vault – Industrial IoT Monitoring Platform (Lead Developer)

- **Led** the end-to-end design and development of a **Javalin and Spring Boot IoT platform** to monitor thousands of industrial refrigeration sensors in real time, reducing manual inspections and preventing costly breakdowns.
- Developed real-time dashboards with live/historical graphs and **WebSocket**-based 1-second exports under 2 seconds; implemented alert mechanisms and optimized **SQLite and InfluxDB pipelines**, achieving 99.9% uptime and millisecond-level query performance.
- Owned architecture decisions and mentored teammates on the IoT platform; collaborated cross-functionally to deliver a scalable, maintainable system.
- **Tech Stack:** Java, Javalin, Spring Boot, SQLite, InfluxDB, WebSocket, JUnit

Project: CloudlyCare – AI-Enabled Healthcare Platform

- Built a production-grade clinical decision support backend in **Python (FastAPI)** with PostgreSQL 16, Redis 7, and pgvector; designed **LangGraph-based multi-agent AI workflows** for context-aware clinical assistance.
- Shipped 553+ automated tests at 95% coverage; used **GitHub Actions CI/CD** and AI-assisted development (Claude, Copilot) to move fast while keeping quality high.
- **Tech Stack:** Python, FastAPI, PostgreSQL, Redis, pgvector, LangGraph, GitHub Actions

Adventure Dhaka Limited, Dhaka

Senior Software Engineer

June 2024 – December 2024

Software Engineer

August 2023 – May 2024

Project: Travel Booking Platform

- Developed sign-up and login modules with **OAuth 2.0 / JWT** authentication and role-based access control, ensuring secure authentication and smooth user onboarding.
- Designed and implemented the hybrid **Localization Module** with dynamic multilingual text and multi-currency support; optimized queries with **Redis caching**, reducing server load, improving response time, and cutting localization costs by up to **40%**.
- Optimized hotel booking by integrating **third-party APIs (Agoda, Expedia, Rakuten)**, caching frequently used data locally, and making API calls on demand, improving system efficiency and reducing operational overhead.
- Managed large-scale data storage and retrieval in **PostgreSQL**, ensuring fast, reliable queries across high-traffic booking flows; mentored junior team members and assisted supervisors in critical architectural decisions.
- **Tech Stack:** Java, Spring Boot, PostgreSQL, MySQL, Redis, Microservices, Kafka

Project: Aircraft Management System

- Developed and maintained a comprehensive **Aircraft Management System** using Java and Spring Boot, streamlining airline resource planning, fleet operations, and day-to-day flight management.
- Designed the **Flight Tracking Module** with real-time status updates and route monitoring, and the **Fuel Management Module** for consumption tracking, refueling schedules, and cost analytics – improving operational efficiency and fuel cost reporting.
- Implemented the **Fleet Monitoring** and **Aircraft Tool Maintenance** modules, ensuring accurate inventory tracking, scheduled maintenance alerts, and improved aircraft availability across the fleet.
- Optimized backend services and complex **MS SQL** queries on large aviation datasets; collaborated on code reviews and architectural discussions for high-quality, maintainable deliverables.
- **Tech Stack:** Java, Spring Boot, MS SQL, REST APIs, Microservices

Project: Conway – Video Conferencing Application

- Developed a video conferencing application (**Conway Project**) using **React and Spring Boot** with a microservices architecture, handling both frontend and backend development.
- Implemented meeting management features (scheduling, history, detailed meeting tracking) and built RESTful APIs for participant management and session history.
- Optimized backend Spring Boot code and real-time session data flow for improved performance; contributed to the web UI and participated in Agile/Scrum code reviews.
- **Tech Stack:** Java, Spring Boot, React, MySQL, Microservices

- Mentored **350+ students** across courses including CSE 173 – Discrete Mathematics, CSE 215 – Object-Oriented Programming, CSE 311 – Database Systems, CSE 327 – Software Engineering, and CSE 482 – Web and Network Technology.
- Created, evaluated, and provided feedback on assignments; assisted faculty with examinations, grading workflows, and course administration to strengthen students’ technical and problem-solving skills.

Education

Bachelor of Computer Science and Engineering (BSc CSE)	CGPA: 3.77/4.00
North South University, Dhaka	January 2016 – May 2020
Higher Secondary Certificate (HSC)	GPA: 5.00/5.00
National Ideal College, Dhaka	2015
Secondary School Certificate (SSC)	GPA: 5.00/5.00
National Ideal School, Dhaka	2013

Project Work

- **AI-Powered Chatbot with RAG:** Built a full-stack AI chatbot using **Spring AI, Groq LLaMA 3.3, and React**. Implemented **Retrieval-Augmented Generation** with PDF ingestion, ONNX embeddings, and a pgvector store for context-aware responses across enterprise knowledge bases.
- **AI Multi-Agent Workflow (LangGraph):** Designed a production multi-agent system that plans, retrieves context, and generates responses – supporting smart follow-ups, summarization, and assistant-style features for enterprise applications.
- **Weather and Rainfall Prediction:** Built a predictive model using machine learning; implemented and compared multiple regression algorithms with Python (scikit-learn) and Weka for data preprocessing, training, and evaluation.
- **Facebook Group Clone Website:** Created a social platform clone with **ReactJS** frontend and **Spring Boot** backend supporting group chats, events, and meetings; optimized API endpoints for smooth real-time interactions.
- **Foodie – Restaurant Management Web App:** Developed a fully responsive restaurant management platform with full CRUD operations; built with ReactJS + Spring Boot, ensuring efficient backend queries.